# Global Difference-Aware Mamba for Hyperspectral Image Change Detection

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***Abstract*—Deep-learning-based methods have been widely applied to hyperspectral image change detection (HSI-CD), where most existing approaches adopt a patchwise framework by taking paired patches from bitemporal HSIs as input and determining whether the central position has changed. However, these methods suffer from several limitations. Being patchwise, they fail to fully exploit the rich information contained in the entire bitemporal HSIs and often overlook useful content outside the training samples. Although some methods incorporate advanced modeling techniques, their receptive fields are still constrained by the patch size, which limits their ability to capture global dependencies. Besides, many approaches underuse the inherent advantage of HSI in capturing subtle spectral differences. To address these challenges, we propose an image-level global difference-aware Mamba (GDAMamba) for HSI-CD, consisting of three key modules: the global contextual difference awareness module (GCDAM), the temporal spectral difference enhancement module (TSDEM), and the adaptive gated fusion module (AGFM). Specifically, GCDAM enables global spatial contextual awareness by establishing associations between each pixel and all others on the difference map. TSDEM processes the bitemporal HSIs in a pixelwise manner, progressively extracts, and enhances spectral differences. These differences are then integrated into the global difference representations through AGFM, which are subsequently fed into a segmentation head to generate the final CD map. Experimental results on three benchmark datasets demonstrate that GDAMamba outperforms several state-of-the-art methods. Our code is available at https://github.com/Preston-Dong/GDAMamba**

***Index Terms*—Global contextual difference awareness, hyperspectral image change detection (HSI-CD), mamba, temporal spectral difference enhancement.**

## I. Introduction

CHANGE detection (CD) using remote sensing data acquired at different temporal phases is critical for monitoring land surface dynamics and has been widely applied in land cover change analysis, agricultural monitoring, and urban planning [1]. Benefiting from detailed spectral sampling across a broad range of wavelengths, hyperspectral images (HSIs) can capture subtle spectral differences [2], making HSI-CD a distinctive and powerful technique in remote sensing CD [3].

Early HSI-CD methods [4], [5] can primarily be categorized into algebra-based [6], transform-based [7], [8], statistical-based [9], classification-based [10] approaches, and their combinations [11]. A representative algebra-based method is change vector analysis (CVA) [6], which detects changes by thresholding the spectral differences. Principal component analysis (PCA) [7] and slow feature analysis (SFA) [8] are widely used in HSI-CD as common transform-based methods. In addition, multivariate alteration detection (MAD) [12] and iteratively reweighted MAD (IR-MAD) [9] leverage canonical correlation analysis (CCA) to find the projection direction that maximizes the differences between bitemporal HSIs. Classification-based methods detect changes by analyzing the difference map or temporal concatenated images using commonly used classifiers, such as $k$-nearest neighbors (KNNs) and support vector machine (SVM) [10]. Despite the significant advancements in traditional methods, they primarily focus on spectral differences between corresponding pixels, lacking effective mechanisms to capture abundant spatial–spectral features, which limits their applicability in complex CD scenarios.

Recently, the advancement of deep learning has significantly accelerated progress in HSI-CD. Various architectures based on convolutional neural network (CNN) [13], [14], transformer [15], and Mamba [16] have also been extensively explored for this task. Thanks to effective localized spatial–spectral feature extraction and parameter sharing scheme, CNNs have played a significant role in HSI-CD. In [13], GETNET is proposed to construct a mixed-affinity matrix through hyperspectral unmixing, which serves as the input for a subsequent 2D-CNN architecture. In [17], a series of UNet models are introduced for HSI-CD, with recurrent residual UNet demonstrating the best performance. Zhao et al. [18] propose a simplified 3-D convolutional autoencoder, which first uses an autoencoder to extract deep spatial–spectral features and then uses these features for classification. Following a similar strategy, Zhang et al. [19] develop a 3-D depthwise separable convolutional autoencoder for HSI-CD. To enhance the representation of finely changed component features at different scales, Luo et al. [20] introduce a multiscale diff-changed feature fusion network (MSDFFN). In [21], the multitemporal symmetric

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fusion network (MTSFNet) is proposed to effectively exploit both difference information and temporal information. Considering the distribution discrepancies between images caused by changes in imaging time, Qin et al. [22] propose the domain alignment dynamic spectral and spatial feature fusion (DADSSFF) model. However, CNN-based methods struggle to model long-range dependencies due to the limitation of local receptive fields, which restricts their ability to comprehensively analyze multitemporal HSIs.

More recently, the ability of transformer to model long-range dependencies and assign dynamic weights has contributed to its growing adoption in HSI-CD. In [23], the joint spectral–spatial–temporal transformer (SSTFormer) is proposed to extract not only spatial information but also crucial spectral and temporal information. In [24], a transformer encoder-based framework named CDFormer is introduced for HSI-CD. Chen et al. [15] present a CNN–transformer-based architecture, TL2GH$^2$T, designed for comprehensive feature representation across spatial, spectral, and spatial–spectral dimensions. Furthermore, Hu et al. [25] design the GlobalMind model to capture interactions among global features, enabling image-level understanding for HSI-CD. Wu et al. [5] propose the differential information enhanced feature exchange network (DIEFEN) to effectively exploit the difference information between HSIs while enhancing the alignment of bitemporal image features. In [26], a dual-branch Siamese spatial–spectral transformer attention network (DBS$^3$TAN) is introduced, which fully leverages CNN and transformer architectures for spatial and spectral feature extraction. To achieve both global and local feature representation, Wang et al. [27] present a global and local attention-based transformer (GLAFormer), which divides attention heads into global and local attention components.

Other state-of-the-art approaches for HSI-CD involve the design and integration of attention mechanisms [28], [29], the utilization of graph neural networks (GNNs) [30], [31], [32], and the application of hyperspectral unmixing techniques [2], [33]. Building on 2D-CNN and self-attention mechanism, Song et al. [28] propose the cross-temporal interaction symmetric attention network (CSANet) to effectively extract and integrate joint spatial–spectral–temporal features. In [29], the HCD-Net framework is proposed, which constructs dual-stream CNNs based on hybrid 2-D and 3-D convolutions, enhanced with 2-D and 3-D squeeze-and-excitation (SE) blocks to extract deep features. To reduce model parameters and computational complexity, Yu et al. [34] develop a gated spectral–spatial–temporal attention network with spectral similarity filtering (HyGSTAN). In [35], an adaptive information weighting and synchronized enhancement network (AIWSEN) is proposed, which integrates an adaptive information weighting attention module, a dual-time synchronic change enhancement module, and a bitemporal image feature selection and fusion module. In addition, Qu et al. [36] introduce graph convolutional network (GCN) into HSI-CD for the first time and present a dual-branch difference amplification GCN (D$^2$AGCN) model. Yang et al. [37] propose a graph-based ARCD, which leverages difference augmentation and progressive reconstruction with limited labels. Based on

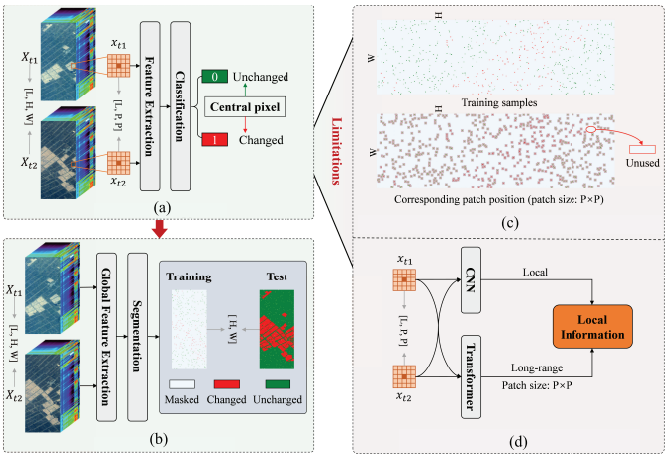



Fig. 1. Comparison between the regular patchwise approach and the image-level approach used in this work. (a) Patchwise approach. (b) Image-level approach. (c) Limitation 1: insufficient utilization of available data. (d) Limitation 2: focus on local information.

hyperspectral unmixing, Hu et al. [38] develop a binary change-guided multiclass change detection network (BCG-Net). Similarly, Xu et al. [39] present an adaptive endmember learning (AEL)-based model for both binary and multiclass CD. Considering the difficulty in obtaining labels for HSI-CD, semi-supervised and self-supervised learning methods have been introduced to achieve reliable CD results using limited or even no labeled data [40]. DCENet [41] is designed for semi-supervised HSI-CD by leveraging a small number of labeled samples alongside a large amount of unlabeled data. In [42], a novel single-temporal mask-based network is proposed, which uses a multiscale mask change simulation strategy to generate pseudo-second-temporal HSIs for training.

As evidenced by previous studies, various effective spatial–spectral–temporal feature extraction techniques have significantly advanced the development of HSI-CD, achieving remarkable success. However, several critical challenges remain to be addressed. As shown in Fig. 1, most HSI-CD methods adopt a patchwise approach, leveraging local spatial–spectral–temporal features from bitemporal HSIs to determine whether the central pixel has changed. At the data level, this patchwise approach does not fully use HSI data, overlooking pixel information not involved in training. At the method level, regardless of network complexity or the use of transformers for long-range dependency modeling, the receptive field remains constrained by the input patch size. Essentially, this still relies on local information rather than true long-range dependency modeling, limiting the ability of deep-learning-based methods to achieve a comprehensive understanding of HSIs. Transformers, which rely on the attention mechanism, have proven effective in modeling long-range dependencies. Building on this, GlobalMind [25] attempts to perform image-level CD. However, the inherently high computational complexity of the attention mechanism presents significant challenges for further advancing image-level HSI-CD research. Furthermore, most methods emphasize spatial–spectral–temporal feature extraction while underusing the high spectral resolution of HSIs. They lack mechanisms

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to amplify spectral differences between bitemporal images, limiting their ability to harness the intrinsic advantages of HSIs.

Benefiting from the ability to capture long-range dependencies with low computational cost, Mamba [43], [44], based on the state-space model (SSM) [45], demonstrates great potential for HSI processing [46], [47], [48], facilitating image-level understanding in HSI-CD [16]. In [46], the MambaHSI model is proposed for image-level HSI classification. Similarly, UNMamba [47] is introduced for blind unmixing, while SSUMamba [48] is developed for HSI denoising. In the HSI-CD field, Wu et al. [49] introduce the spatial–spectral Mamba interactive fusion (SSMIF) network, while Li et al. [16] propose the GASSM model by integrating Mamba with global attention. However, both of them remain patchwise and have yet to fully harness the strength of Mamba in modeling long-range dependencies for image-level CD. To fully leverage all available HSI data and develop an effective CD framework capable of capturing global contextual dependencies, we propose the image-level global difference-aware Mamba (GDAMamba), the first fully Mamba-based architecture specifically designed for image-level HSI-CD. GDAMamba extracts global difference information through stacked encoder blocks, each comprising three key components: a global contextual difference awareness module (GCDAM) for establishing associations between each pixel and all others on the difference map, a temporal spectral difference enhancement module (TSDEM) for amplifying pixelwise spectral differences, and an adaptive gated fusion module (AGFM) for dynamically integrating enhanced differences into global contextual representations. The main contributions of this article are as follows.

1) GDAMamba is the first Mamba-based image-level HSI-CD method, where the discriminative features at each position are derived from the entire bitemporal image pair, establishing global difference dependencies for enhanced CD while maintaining strong performance even with limited training samples.
2) The GCDAM is designed to capture global contextual difference information at the spatial dimensions, enabling image-level difference awareness and enhancing the understanding of bitemporal HSIs, thereby improving the distinction between changed and unchanged regions.
3) The TSDEM is proposed to perform pixelwise comparisons of the bitemporal spectra using a weight-shared SSM, effectively capturing and enhancing subtle spectral differences, which are then integrated via AGFM to refine the global difference representations and improve detection accuracy.

The rest of this article is organized as follows. Section II provides a detailed description of the proposed GDAMamba. Section III presents extensive experiments on three datasets. Section IV gives a discussion, including ablation study, parameter analysis, and comparative analysis. Section V concludes this article.

## II. Methodology

This section starts with an overview of the proposed GDAMamba, followed by a detailed explanation of its key components. Then, the training strategy and loss function are introduced. Finally, a patchwise variant, GDAMamba-P, which shares the same structure as GDAMamba, is also presented.

### A. Overview of the Proposed GDAMamba

The framework of the proposed GDAMamba is illustrated in Fig. 2. Instead of a patchwise approach, it directly processes the bitemporal HSIs as the input and generates CD results through segmentation. GDAMamba comprises three identical encoder blocks, each integrating three key components: a GCDAM, a TSDEM, and an AGFM. Given the input bitemporal HSIs $\boldsymbol{X}_{t1} \in \mathbb{R}^{L\times H\times W}$ and $\boldsymbol{X}_{t2} \in \mathbb{R}^{L\times H\times W}$, the difference map $\boldsymbol{D} \in \mathbb{R}^{L\times H\times W}$ is first computed through elementwise subtraction

$$\boldsymbol{D} = \boldsymbol{X}_{t2} - \boldsymbol{X}_{t1}. \tag{1}$$

Subsequently, $\boldsymbol{D}$ is embedded and processed through one convolutional layer with $1 \times 1$ kernel size to extract feature representations, yielding $\boldsymbol{F}_D^{(0)} \in \mathbb{R}^{C\times H\times W}$

$$\boldsymbol{F}_D^{(0)} = f_{\text{Conv1}}^{1\times 1}(\boldsymbol{D}) \tag{2}$$

which serves as the initial input to GCDAM in the first encoder block. Simultaneously, $\boldsymbol{X}_{t1}$ and $\boldsymbol{X}_{t2}$ are fed into a convolutional layer with shared weights to derive image feature representations $\boldsymbol{F}_{t1}^{(0)}$ and $\boldsymbol{F}_{t2}^{(0)}$, capturing essential features from different temporal phases

$$\begin{aligned}\boldsymbol{F}_{t1}^{(0)} &= f_{\text{Conv2}}^{1\times 1}(\boldsymbol{X}_{t1})\\ \boldsymbol{F}_{t2}^{(0)} &= f_{\text{Conv2}}^{1\times 1}(\boldsymbol{X}_{t2})\end{aligned} \tag{3}$$

which are processed in a pixelwise manner along the temporal dimension and serve as the input to the TSDEM in the first encoder block. Subsequently, the temporal spectral feature differences are integrated into the global contextual difference features, and the final global aware difference representations $\boldsymbol{F}_D^{(3)}$ are fed into the segmentation head to generate the prediction results $\hat{\boldsymbol{Y}} \in \mathbb{R}^{2\times H\times W}$

$$\hat{\boldsymbol{Y}} = \text{Softmax}\left(f_{\text{Conv3}}^{1\times 1}\left(\boldsymbol{F}_D^{(3)}\right)\right). \tag{4}$$

Subsequently, the CD map is obtained by selecting the index corresponding to the maximum predicted probability for each pixel.

### B. Global Contextual Difference Awareness Module

To achieve long-range dependency modeling, ensuring that the change information of each pixel is derived from all the pixels in the entire difference map for global difference awareness, the VMamba-based GCDAM is proposed. It establishes pixelwise associations through 2D-selective-scan (SS2D) [44] and the S6 block [43], as shown in Fig. 3. GCDAM constructs global dependencies between each pixel feature and all other pixel features along four distinct scanning paths and comprises three branches: the SS2D branch for long-range dependency modeling, the multiplication branch for feature interaction, and

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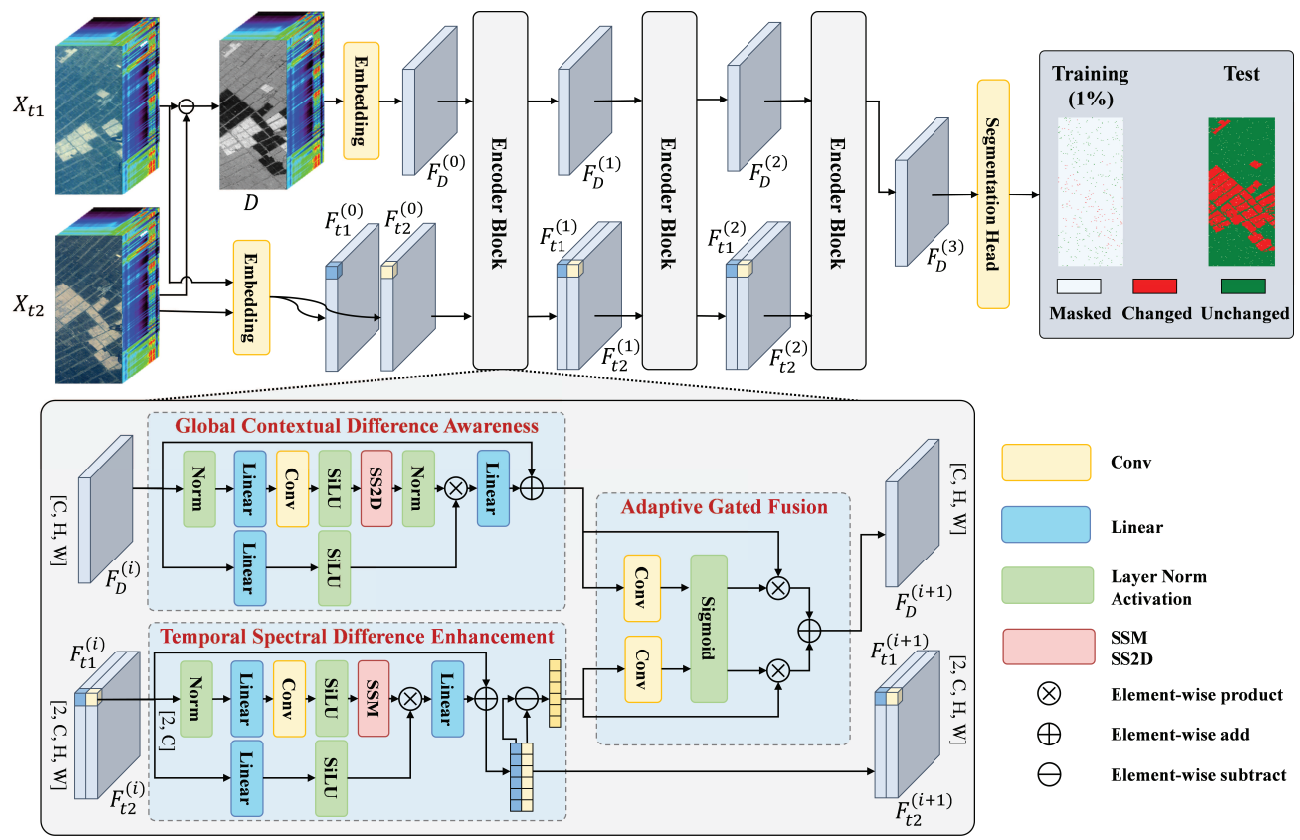



Fig. 2. Overview of the proposed GDAMamba, which takes the bitemporal HSIs as the input to generate CD results through segmentation. It consists of three identical encoder blocks, each incorporating three essential components: GCDAM for capturing global contextual difference features, TSDEM for enhancing temporal spectral differences, and AGFM for integrating spectral differences into global difference representations.

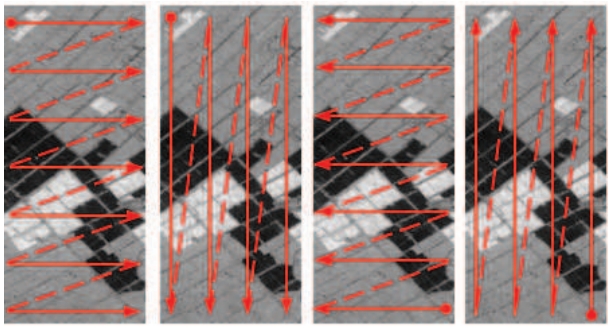


Fig. 3. Illustration of SS2D for modeling long-range dependencies in difference features $\boldsymbol{F}_D^{(i)}$.

the residual branch for information preservation. Assuming that the input to the GCDAM of the $(i+1)$th encoder block is $\boldsymbol{F}_D^{(i)}$ and the output is $\boldsymbol{T}_D^{(i+1)}$, the SS2D branch can be expressed as follows:

$$\begin{aligned}\boldsymbol{T}_{D1}^{(i+1)} &= \mathrm{SiLU}\left(f_{\mathrm{Conv}}\left(f_{\mathrm{Linear}}\left(\mathrm{LN}\left(\boldsymbol{F}_D^{(i)}\right)\right)\right)\right)\\ \boldsymbol{T}_{D2}^{(i+1)} &= \mathrm{LN}\left(\mathrm{SS2D}\left(\boldsymbol{T}_{D1}^{(i+1)}\right)\right)\end{aligned} \tag{5}$$

where $\boldsymbol{T}_{D1}^{(i+1)}$ denotes the transformed features used as input to SS2D, while $\boldsymbol{T}_{D2}^{(i+1)}$ represents the features after constructing global dependencies. In addition, $f_{\mathrm{Conv}}$, $f_{\mathrm{Linear}}$, SiLU, and LN correspond to the convolution operation, fully connected layer, SiLU activation function, and layer normalization, respectively. After that, the output of the combined multiplication branch can be expressed as follows:

$$\begin{aligned}\boldsymbol{T}_{D3}^{(i+1)} &= \mathrm{SiLU}\left(f_{\mathrm{Linear}}\left(\boldsymbol{F}_D^{(i)}\right)\right)\\ \boldsymbol{T}_{D4}^{(i+1)} &= \boldsymbol{T}_{D2}^{(i+1)}\boldsymbol{T}_{D3}^{(i+1)}\end{aligned} \tag{6}$$

where $\boldsymbol{T}_{D3}^{(i+1)}$ is the result of the multiplication branch, which is subsequently fused to produce $\boldsymbol{T}_{D4}^{(i+1)}$, representing the globally interacted features. Finally, the residual branch serves to preserve the initial input information, effectively preventing overfitting

$$\boldsymbol{T}_D^{(i+1)} = \boldsymbol{F}_D^{(i)} + f_{\mathrm{Linear}}\left(\boldsymbol{T}_{D4}^{(i+1)}\right) \tag{7}$$

which represents the global contextual difference features captured along the spatial dimensions, serving as the spatial-domain input for the subsequent feature fusion process.

### C. Temporal Spectral Difference Enhancement Module

TSDEM is designed to fully exploit the spectral characteristics of HSIs to detect subtle changes between bitemporal spectra and progressively enhance them for a more accurate representation of changes. It shares the same overall structure as GCDAM but processes spectral pairs at corresponding pixel positions using a weight-shared SSM in a pixelwise approach. Let the inputs to TSDEM of the $(i+1)$th encoder block be

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the bitemporal spectral features $\boldsymbol{F}_{t1}^{(i)}$ and $\boldsymbol{F}_{t2}^{(i)}$; they are first concatenated along the channel dimension and then flattened

$$\boldsymbol{T}_{S1}^{(i+1)} = \text{Flatten}\left(\text{Cat}\left(\boldsymbol{F}_{t1}^{(i)}, \boldsymbol{F}_{t2}^{(i)}\right)\right) \tag{8}$$

where $\boldsymbol{T}_{S1}^{(i+1)} \in \mathbb{R}^{H\cdot W\times 2\times C}$ denotes the temporal spectral features after concatenation and flattening. Subsequently, the SSM branch can be formulated as follows:

$$\begin{aligned}\boldsymbol{T}_{S2}^{(i+1)} &= \text{SiLU}\left(f_{\text{Conv}}\left(f_{\text{Linear}}\left(\text{LN}\left(\boldsymbol{T}_{S1}^{(i+1)}\right)\right)\right)\right)\\ \boldsymbol{T}_{S3}^{(i+1)} &= \text{SSM}\left(\boldsymbol{T}_{S2}^{(i+1)}\right)\end{aligned} \tag{9}$$

where $\boldsymbol{T}_{S3}^{(i+1)}$ denotes the output of the SSM branch. Subsequent feature interaction with the multiplication branch and the residual branch refines the features while preserving the original information, resulting in the final bitemporal pixel-level spectral difference-aware features $\boldsymbol{T}_{S6}^{(i+1)}$

$$\begin{aligned}\boldsymbol{T}_{S4}^{(i+1)} &= \text{SiLU}\left(f_{\text{Linear}}\left(\boldsymbol{T}_{S1}^{(i+1)}\right)\right)\\ \boldsymbol{T}_{S5}^{(i+1)} &= \boldsymbol{T}_{S3}^{(i+1)}\boldsymbol{T}_{S4}^{(i+1)}\\ \boldsymbol{T}_{S6}^{(i+1)} &= \boldsymbol{T}_{S1}^{(i+1)} + f_{\text{Linear}}\left(\boldsymbol{T}_{S5}^{(i+1)}\right).\end{aligned} \tag{10}$$

Subsequently, $\boldsymbol{T}_{S6}^{(i+1)}$ is split along the channel dimension and reshaped to the original image format to obtain $\boldsymbol{F}_{t1}^{(i+1)}$ and $\boldsymbol{F}_{t2}^{(i+1)}$ after temporal interaction

$$\begin{aligned}\boldsymbol{F}_{t1}^{(i+1)} &= \text{Reshape}\left(\boldsymbol{T}_{S6}^{(i+1)}[:,0,:]\right)\\ \boldsymbol{F}_{t2}^{(i+1)} &= \text{Reshape}\left(\boldsymbol{T}_{S6}^{(i+1)}[:,1,:]\right)\end{aligned} \tag{11}$$

where their difference $\boldsymbol{F}_{\text{td}}^{(i+1)}$ is then computed to serve as the spectral difference enhancement features for the TSDEM in the $(i+1)$th encoder block

$$\boldsymbol{F}_{\text{td}}^{(i+1)} = \boldsymbol{F}_{t2}^{(i+1)} - \boldsymbol{F}_{t1}^{(i+1)} \tag{12}$$

which acts as the spectral-domain input for the following feature fusion stage.

### D. Adaptive Gated Fusion Module

AGFM aims to integrate spectral differences into global spatial contextual difference features to enhance their discriminative capacity. In this article, AGFM is implemented as a lightweight adaptive gating mechanism, which generates weight masks for $\boldsymbol{T}_{D}^{(i+1)}$ and $\boldsymbol{F}_{\text{td}}^{(i+1)}$ via convolutional layers, followed by additive fusion of the reweighted features:

$$\begin{aligned}\boldsymbol{M}_{D}^{(i+1)} &= \text{Sigmoid}\left(f_{\text{Conv}}\left(\boldsymbol{T}_{D}^{(i+1)}\right)\right)\\ \boldsymbol{M}_{\text{td}}^{(i+1)} &= \text{Sigmoid}\left(f_{\text{Conv}}\left(\boldsymbol{F}_{\text{td}}^{(i+1)}\right)\right)\\ \boldsymbol{F}_{D}^{(i+1)} &= \boldsymbol{M}_{D}^{(i+1)}\boldsymbol{T}_{D}^{(i+1)} + \boldsymbol{M}_{\text{td}}^{(i+1)}\boldsymbol{F}_{\text{td}}^{(i+1)}\end{aligned} \tag{13}$$

where $\boldsymbol{M}_{D}^{(i+1)}$ and $\boldsymbol{M}_{\text{td}}^{(i+1)}$ denote the weight masks corresponding to $\boldsymbol{T}_{D}^{(i+1)}$ and $\boldsymbol{F}_{\text{td}}^{(i+1)}$, respectively. The fused output $\boldsymbol{F}_{D}^{(i+1)}$ serves as the final representation of the $(i+1)$th encoder block. In this article, three such encoder blocks are used, and the global contextual difference-aware features $\boldsymbol{F}_{D}^{(3)}$, progressively refined through spectral-domain enhancement, are fed into the segmentation head to generate the final CD results.

**Algorithm 1** Training and Prediction Procedures of GDAMamba

TRAIN

**Input:** Bitemporal HSIs $\boldsymbol{X}_{t1}$ and $\boldsymbol{X}_{t2}$, and one-hot encoded GT $\boldsymbol{Y}$

**Initialize:**

Initialize network parameters $\Theta$ with random values

Generate training mask $\boldsymbol{m}$

$\boldsymbol{X}_{t1} \leftarrow (\boldsymbol{X}_{t1} - \min(\boldsymbol{X}_{t1}))/(\max(\boldsymbol{X}_{t1}) - \min(\boldsymbol{X}_{t1}))$

$\boldsymbol{X}_{t2} \leftarrow (\boldsymbol{X}_{t2} - \min(\boldsymbol{X}_{t2}))/(\max(\boldsymbol{X}_{t2}) - \min(\boldsymbol{X}_{t2}))$

**For** $e < E$ **do**

$\boldsymbol{F}_{D}^{(0)} \leftarrow f_{\text{Conv1}}^{1\times1}(\boldsymbol{X}_{t2} - \boldsymbol{X}_{t1})$

$\boldsymbol{F}_{t1}^{(0)} \leftarrow f_{\text{Conv2}}^{1\times1}(\boldsymbol{X}_{t1})$

$\boldsymbol{F}_{t2}^{(0)} \leftarrow f_{\text{Conv2}}^{1\times1}(\boldsymbol{X}_{t2})$

**For** $i = 0, 1, \ldots, I-1$ **do**

$\boldsymbol{T}_{D}^{(i+1)} \leftarrow \text{GCDAM}(\boldsymbol{F}_{D}^{(i)})$

$\boldsymbol{F}_{t1}^{(i+1)}, \boldsymbol{F}_{t2}^{(i+1)} \leftarrow \text{TSDEM}(\boldsymbol{F}_{t1}^{(i)}, \boldsymbol{F}_{t2}^{(i)})$

$\boldsymbol{F}_{\text{td}}^{(i+1)} \leftarrow \boldsymbol{F}_{t2}^{(i+1)} - \boldsymbol{F}_{t1}^{(i+1)}$

$\boldsymbol{F}_{D}^{(i+1)} \leftarrow \text{AGFM}(\boldsymbol{T}_{D}^{(i+1)}, \boldsymbol{F}_{\text{td}}^{(i+1)})$

**end**

$\hat{\boldsymbol{Y}}_{t} \leftarrow \text{Softmax}(f_{\text{Conv3}}^{1\times1}(\boldsymbol{F}_{D}^{(I)}))$

$\mathcal{L} \leftarrow$ (14)

Update the parameters $\Theta$ with Adam optimizer

**end**

**Output:** Network parameters $\Theta$

PREDICT

**Input:** Bitemporal HSIs $\boldsymbol{X}_{t1}$ and $\boldsymbol{X}_{t2}$

1) Load trained network parameters $\Theta$
2) generate change probability map $\hat{\boldsymbol{Y}}$

**Output:** $\hat{\boldsymbol{Y}}$

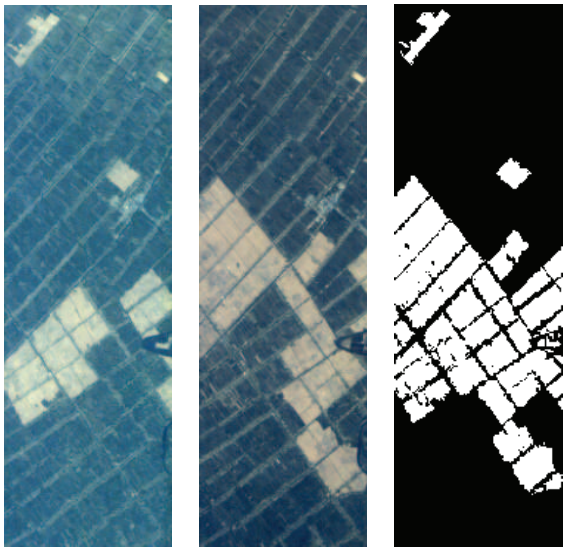
### E. Training and Loss Function

GDAMamba is an image-level CD method designed to achieve global contextual difference awareness. This also implies that unlike most conventional methods, CD results of GDAMamba are obtained through a segmentation paradigm. Consequently, both the training strategy and loss function differ substantially from the patchwise approaches. In this article, 1% of the ground-truth labels are randomly sampled for training, and a corresponding training mask is generated simultaneously, as illustrated in Fig. 2. The remaining pixels are excluded from loss computation using a masking mechanism. Accordingly, a masked cross-entropy loss function is used during training

$$\mathcal{L} = -\frac{1}{\sum_{(h,w)}\boldsymbol{m}^{(h,w)}}\sum_{h=1}^{H}\sum_{w=1}^{W}\boldsymbol{m}^{(h,w)}\sum_{c=1}^{2}\boldsymbol{Y}_{c}^{(h,w)}\log\left(\hat{\boldsymbol{Y}}_{c}^{(h,w)}\right) \tag{14}$$

where $\boldsymbol{Y}$ denotes the one-hot encoded ground truth, $\hat{\boldsymbol{Y}}$ represents the predicted probability map generated by the GDAMamba model, and $\boldsymbol{m} \in \mathbb{R}^{H\times W}$ is the training mask, where pixels not involved in training are assigned a value of 0, and those selected for training are assigned a value of 1. The overall training and prediction procedures of our GDAMamba are outlined in Algorithm 1.

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(a) (b) (c)

Fig. 4. Farmland dataset. (a) HSI was acquired on May 3, 2006. (b) HSI was acquired on April 23, 2007. (c) GT.

### F. Regular Patchwise GDAMamba-P

For a fair comparison to conventional patchwise methods, a patchwise variant named GDAMamba-P is constructed based on the same architecture of GDAMamba. It takes a $5 \times 5$ window-sized patch as input and extracts local contextual difference features from the bitemporal HSIs to determine the class of the central pixel. Assuming the input bitemporal HSI patch pair is denoted as $\boldsymbol{x}_{t1} \in \mathbb{R}^{L\times5\times5}$ and $\boldsymbol{x}_{t2} \in \mathbb{R}^{L\times5\times5}$, the resulting center pixel features $\boldsymbol{f}_{\mathrm{dc}}^{(3)} \in \mathbb{R}^{C}$ can be formulated as follows:

$$\begin{aligned} \boldsymbol{F}_{d}^{(3)} &= \mathrm{GDAMamba}\left(\boldsymbol{x}_{t1}, \boldsymbol{x}_{t2}\right) \\ \boldsymbol{f}_{\mathrm{dc}}^{(3)} &= \boldsymbol{F}_{d}^{(3)}[:, 2, 2] \end{aligned} \tag{15}$$

where $\boldsymbol{F}_{d}^{(3)}$ denotes the features extracted from the bitemporal patch pair, and $\boldsymbol{f}_{\mathrm{dc}}^{(3)}$ represents the difference features corresponding to the central pixel. The CD result is then obtained by feeding $\boldsymbol{f}_{\mathrm{dc}}^{(3)}$ into (4).

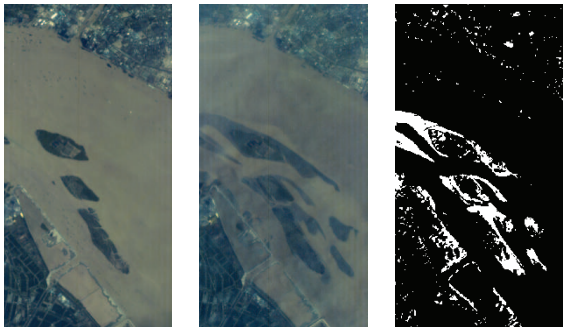
## III. Experimental Results

In this section, we first introduce the three benchmark datasets and the five evaluation metrics used. Then, we elaborate the comparison methods and experimental settings. Finally, quantitative results and visual comparisons of GDAMamba and its patchwise variant GDAMamba-P against other state-of-the-art methods are provided.

### A. Datasets and Evaluation Metrics

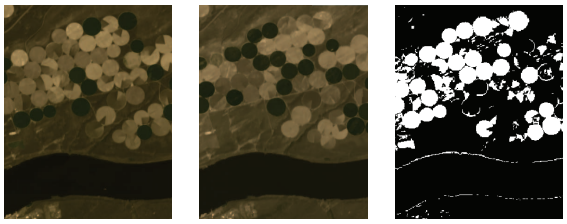
*1) Datasets:* Three benchmark datasets, i.e., Farmland dataset [50], River dataset [13], and USA dataset [50], are used to evaluate the feasibility and effectiveness of the proposed GDAMamba and its variant.

The first dataset, shown in Fig. 4, is located in Yancheng, Jiangsu Province, China, and was acquired on May 3, 2006 and April 23, 2007 by the Earth Observing-1 (EO-1) Hyperion sensor. Each image contains $450 \times 140$ pixels with a spatial resolution of 30 m and covers spectral bands ranging from 0.4 to 2.5 $\mu$m. After removing noise and water absorption bands, 155 spectral bands are retained for the experiment.



(a) (b) (c)

Fig. 5. River dataset. (a) HSI was acquired on May 3, 2013. (b) HSI was acquired on December 31, 2013. (c) GT.



(a) (b) (c)

Fig. 6. USA dataset. (a) HSI was acquired on May 1, 2004. (b) HSI was acquired on May 8, 2007. (c) GT.

The second dataset, displayed in Fig. 5, was captured over a riverine region in Jiangsu Province, China, on May 3, 2013 and December 31, 2013, using the EO-1 Hyperion sensor. Each image has a spatial size of $463 \times 241$ pixels with a spatial resolution of 30 m and a spectral resolution of 10 nm. After eliminating noisy bands, 198 spectral bands are retained for validation.

The third dataset, illustrated in Fig. 6, covers an irrigated agricultural area in Hermiston City, USA, and was acquired by the EO-1 Hyperion sensor on May 1, 2004 and May 8, 2007. Each image consists of $307 \times 241$ pixels and contains 154 spectral bands.

*2) Evaluation Metrics:* To comprehensively and quantitatively evaluate the CD performance of different methods, five metrics are adopted in this article: overall accuracy (OA), Kappa coefficient (Kappa), precision ($P$), recall ($R$), and $F1$-score ($F1$). Their definitions are as follows:

$$\begin{aligned} \mathrm{OA} &= \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}, \quad P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ R &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{TN}}, \quad \mathrm{F1} = \frac{2P\times R}{P+R}, \quad \mathrm{Kappa} = \frac{\mathrm{OA}-p_e}{1-p_e} \\ p_e &= \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \end{aligned} \tag{16}$$

where true positive (TP) denotes the number of pixels labeled as change and correctly detected as change by the algorithm, false positive (FP) denotes the number of unchanged pixels

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that are incorrectly detected as change, true negative (TN) represents the number of unchanged pixels correctly detected as unchanged, and false negative (FN) refers to the number of changed pixels that are incorrectly detected as unchanged.

### B. Compared Methods and Experimental Settings

The comparison methods adopted in this article primarily include recent competitive approaches, such as CSANet [28], SSTFormer [23], MSDFFN [20], DIEFEN [5], HyGSTAN [34], and several state-of-the-art methods including GLAFormer [27], DADSSFF [22], AIWSEN [35], and GASSM [16], and traditional methods such as CVA [6] and DPCA [7].

For the individual deep learning methods, the experimental settings are kept consistent with those reported in their original papers. Specifically, the input patch size is set to $9 \times 9$ for CSANet, MSDFFN, and GLAFormer, $7 \times 7$ for DADSSFF, AIWSEN, and GASSM, and $5 \times 5$ for SSTFormer, DIEFEN, HyGSTAN, and the proposed GDAMamba-P. In contrast, the proposed GDAMamba is an image-level CD method that directly takes bitemporal HSIs as input. For both the GDAMamba and GDAMamba-P models, a total of three encoder blocks are used, with the number of feature channels $C$ set to 32.

For all the deep learning methods, the number of training epochs is set to 100. The SGD optimizer with a momentum of 0.9 is adopted for all non-Mamba-based models, while the Adam optimizer is used for Mamba-based models including GASSM, as well as the proposed GDAMamba and GDAMamba-P. The initial learning rate is uniformly set to $5e^{-3}$ and decays by a factor of 0.1 every 35 epochs. For all the patchwise approaches, the batch size is set to 64. All the models are implemented using the PyTorch framework and are trained and evaluated on an NVIDIA RTX 4090 GPU.

In the experiments, 1% of the ground-truth labels are randomly selected as training samples (with results under different proportions discussed in Section IV). To ensure evaluation reliability, each method is independently run five times using different random seeds, and the mean and standard deviation of the results are reported.

### C. Experimental Results

Table I presents the CD results on the three benchmark datasets using 1% of the ground truth as training samples. The best performance among all the methods is highlighted in bold and marked with ①, while the second-best performance is indicated with ②. Due to the class imbalance caused by the relatively small proportion of changed regions, OA and Kappa are jointly considered to evaluate overall performance. Furthermore, the $F1$-score, computed as the harmonic mean of $P$ and $R$, is used for comprehensive assessment, with $P$ and $R$ also analyzed individually for finer grained interpretation. Compared with traditional methods such as CVA and DPCA, deep-learning-based approaches achieve further improvements in quantitative metrics.

On the Farmland dataset, the proposed GDAMamba achieves the best performance in terms of OA, Kappa, $F1$-score, and $P$. Although it is slightly weaker than DIEFEN in terms of $R$, it still surpasses all other methods. Meanwhile, GDAMamba-P, though weaker than the globally difference-aware image-level GDAMamba, consistently outperforms other patchwise models. Specifically, GDAMamba achieves an OA of 98.38% and a Kappa coefficient of 0.9608, along with a leading $F1$-score of 97.22%. The high OA and Kappa values suggest that the proposed GDAMamba effectively mitigates the impact of sample imbalance on the Farmland dataset and yields predictions that are highly consistent with the ground truth. Moreover, the superior $F1$-score reflects its strong capability in capturing subtle changes across bitemporal HSIs. In addition, AIWSEN also demonstrates competitive CD performance, which can be attributed to its use of information entropy to enhance change feature extraction and its interactive information modeling between bitemporal HSI patch pairs.

Fig. 7 presents the CD maps produced by different methods on the Farmland dataset, where white indicates TPs, black represents TNs, red corresponds to false alarms (FP), and blue denotes missed detections (FN). Compared with traditional methods and patchwise deep learning approaches, the CD map generated by the proposed GDAMamba exhibits more continuous change regions with sharper boundaries. Overall, it yields fewer false alarms and missed detections, along with reduced scattered noise, demonstrating the model's strong capability in detecting subtle changes and distinguishing between change and no-change regions on this dataset.

On the River dataset, GDAMamba achieves the best CD performance, with OA, Kappa, and $F1$-score reaching 97.49%, 0.8412%, and 85.50%, respectively, outperforming all other deep learning methods. In terms of precision and recall, GDAMamba-P achieves the highest precision, while GDAMamba obtains higher recall compared with GDAMamba-P. Although the highest recall is observed in traditional methods, their precision is significantly lower, indicating a higher rate of false alarms. As $F1$-score represents the harmonic mean of precision and recall, the superior $F1$-score of GDAMamba suggests a well-balanced tradeoff between the two, demonstrating its robust and reliable CD performance. Fig. 8 presents the visualization results of each method on the River dataset. It is observed that traditional methods such as CVA and DPCA produce a larger number of false alarms, as indicated by the extensive red regions, which is consistent with the previously discussed quantitative metrics. In contrast, deep-learning-based methods demonstrate a significant reduction in false alarms. Notably, the proposed GDAMamba yields change regions that are more consistent with the ground truth and exhibit clearer boundaries, reflecting its superior capability in accurate and precise CD.

On the USA dataset, GDAMamba trained with only 1% of the labels still achieves the best performance in terms of OA (95.88%), Kappa (0.8823), and $F1$-score (90.88%), with $F1$-score surpassing 90% for the first time. In terms of precision and recall, the precision values of deep learning methods are generally lower than those of traditional methods, while their recall values are significantly higher, with GDAMamba achieving the highest recall among all the methods. This suggests that traditional methods are more prone to missed

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TABLE I
PERFORMANCE COMPARISON ON THE FARMLAND, RIVER, AND USA DATASETS WITH 1% LABELED TRAINING SAMPLES

| Datasets | Methods | OA (%) | Kappa (×100) | F1 (%) | P (%) | R (%) |
|---|---|---|---|---|---|---|
| Farmland | CVA | 95.67 | 89.60 | 92.66 | 91.24 | 94.12 |
| | DPCA | 95.49 | 89.36 | 92.60 | 88.33 | 97.30 |
| | CSANet [2022] | 96.63±0.46 | 91.82±1.13 | 94.19±0.80 | 94.39±0.85 | 93.98±0.97 |
| | SSTFormer [2022] | 96.29±0.17 | 91.07±0.43 | 93.69±0.31 | 92.55±1.12 | 94.89±1.35 |
| | MSDFFN [2023] | 96.85±0.37 | 92.36±0.93 | 94.59±0.68 | 94.13±1.16 | 95.09±1.85 |
| | DIEFEN [2024] | 96.23±0.35 | 91.07±0.79 | 93.76±0.54 | 90.38±1.11 | **97.41±0.48**① |
| | HyGSTAN [2024] | 96.48±0.42 | 91.29±1.17 | 93.73±0.90 | 96.73±1.53② | 91.01±2.97 |
| | GLAFormer [2025] | 97.03±0.30 | 92.78±0.74 | 94.87±0.53 | 94.95±1.09 | 94.82±1.35 |
| | DADSSFF [2025] | 96.56±0.39 | 91.52±1.01 | 93.92±0.76 | 96.37±0.60 | 91.61±1.78 |
| | AIWSEN [2025] | 97.72±0.18 | 94.49±0.44 | 96.11±0.31 | 95.16±0.46 | 97.08±0.68 |
| | GASSM [2025] | 97.61±0.11 | 94.21±0.28 | 95.88±0.21 | 96.09±0.62 | 95.69±0.82 |
| | **GDAMamba-P** | 98.02±0.21② | 95.19±0.50② | 96.58±0.35② | 96.68±0.94 | 96.49±0.44 |
| | **GDAMamba** | **98.38±0.17**① | **96.08±0.41**① | **97.22±0.29**① | **97.10±0.60**① | 97.34±0.29② |
| River | CVA | 92.68 | 65.74 | 69.54 | 54.47 | 96.13② |
| | DPCA | 90.02 | 57.85 | 62.80 | 46.45 | **96.94**① |
| | CSANet [2022] | 96.15±0.13 | 74.49±1.05 | 76.58±1.00 | 81.32±2.14 | 72.47±2.67 |
| | SSTFormer [2022] | 92.85±0.59 | 48.76±7.45 | 52.50±7.33 | 61.43±4.09 | 46.32±8.76 |
| | MSDFFN [2023] | 96.09±0.15 | 74.11±0.75 | 76.23±0.68 | 81.04±2.11 | 72.03±1.42 |
| | DIEFEN [2024] | 95.91±0.20 | 73.14±1.68 | 75.36±1.58 | 79.03±1.77 | 72.13±3.14 |
| | HyGSTAN [2024] | 92.87±0.73 | 58.68±2.77 | 62.58±2.39 | 57.89±4.11 | 68.32±1.58 |
| | GLAFormer [2025] | 96.78±0.13 | 78.85±1.36 | 80.60±1.31 | 84.60±1.32 | 77.09±3.46 |
| | DADSSFF [2025] | 96.74±0.08 | 78.79±0.69 | 80.57±0.71 | 83.76±3.41 | 77.95±4.28 |
| | AIWSEN [2025] | 96.84±0.15 | 79.23±1.08 | 80.95±1.01 | 84.99±1.95 | 77.37±2.39 |
| | GASSM [2025] | 97.08±0.07 | 80.94±0.56 | 82.53±0.53 | 86.09±1.55② | 79.32±1.95 |
| | **GDAMamba-P** | 97.42±0.11② | 83.33±0.70② | 84.74±0.64② | **87.46±1.67**① | 82.22±1.47 |
| | **GDAMamba** | **97.49±0.06**① | **84.12±0.29**① | **85.50±0.26**① | 85.82±1.11 | 85.20±0.85 |
| USA | CVA | 92.00 | 74.12 | 78.83 | 97.72② | 66.06 |
| | DPCA | 92.47 | 75.80 | 80.26 | **98.16**① | 67.88 |
| | CSANet [2022] | 94.55±0.25 | 84.05±0.69 | 87.53±0.54 | 90.32±1.62 | 84.97±1.54 |
| | SSTFormer [2022] | 94.27±0.25 | 83.23±0.58 | 86.89±0.43 | 89.88±2.33 | 84.18±1.87 |
| | MSDFFN [2023] | 94.61±0.30 | 84.37±0.80 | 87.83±0.60 | 89.38±1.65 | 86.37±1.24 |
| | DIEFEN [2024] | 93.56±0.35 | 82.15±0.94 | 86.35±0.71 | 82.69±0.93 | 90.36±0.62② |
| | HyGSTAN [2024] | 90.23±1.75 | 72.52±3.60 | 78.84±2.38 | 78.37±7.26 | 80.02±3.54 |
| | GLAFormer [2025] | 94.21±0.24 | 83.28±0.64 | 87.01±0.48 | 88.04±1.36 | 86.03±1.01 |
| | DADSSFF [2025] | 95.08±0.19 | 85.39±0.64 | 88.50±0.55 | 93.56±2.09 | 84.06±2.31 |
| | AIWSEN [2025] | 95.37±0.44 | 86.71±1.14 | 89.70±0.85 | 90.04±1.92 | 89.40±1.00 |
| | GASSM [2025] | 95.55±0.37② | 87.06±0.96② | 89.91±0.71② | 92.10±1.94 | 87.86±0.88 |
| | **GDAMamba-P** | 94.93±0.39 | 85.42±1.26 | 88.69±1.02 | 89.02±0.75 | 88.42±2.36 |
| | **GDAMamba** | **95.88±0.21**① | **88.23±0.52**① | **90.88±0.38**① | 90.71±1.21 | **91.07±0.69**① |

detections. The superior $F1$-score further demonstrates that GDAMamba achieves a well-balanced performance between precision and recall on this dataset as well. The CD maps produced by each method on this dataset are illustrated in Fig. 9. It can be observed that the result of GDAMamba is significantly clearer, with more detailed and well-defined edges. Moreover, it successfully identifies most of the change regions that are missed by other methods. This highlights GDAMamba's superior ability to capture subtle and small-scale changes with high spatial consistency.

On all the three benchmark datasets, GDAMamba consistently achieves the highest scores in terms of OA, Kappa, and $F1$-score. The high OA and Kappa values demonstrate the superiority of the proposed algorithm in feature extraction and its effectiveness in alleviating the class imbalance between changed and unchanged regions. Meanwhile, the high $F1$-score indicates that GDAMamba strikes a favorable balance between false alarms and missed detections. This suggests that the model not only exhibits strong sensitivity to subtle changes but also effectively suppresses interference from unchanged regions. Furthermore, when compared with its patchwise variant GDAMamba-P, GDAMamba consistently outperforms it, highlighting the benefits of global contextual long-range dependency modeling. These results validate the effectiveness of the proposed approach in both architectural design and global information utilization for HSI-CD.

## IV. DISCUSSION

This section first presents the experimental results under varying training sample ratios and offers a detailed analysis of the computational complexity of each model. Subsequently, an ablation study is conducted to evaluate the contributions of individual modules, followed by an in-depth analysis of GDAMamba and GDAMamba-P.

### A. Impact of Training Sample Ratio

Fig. 10 illustrates the CD performance of different methods under varying training sample ratios on the three

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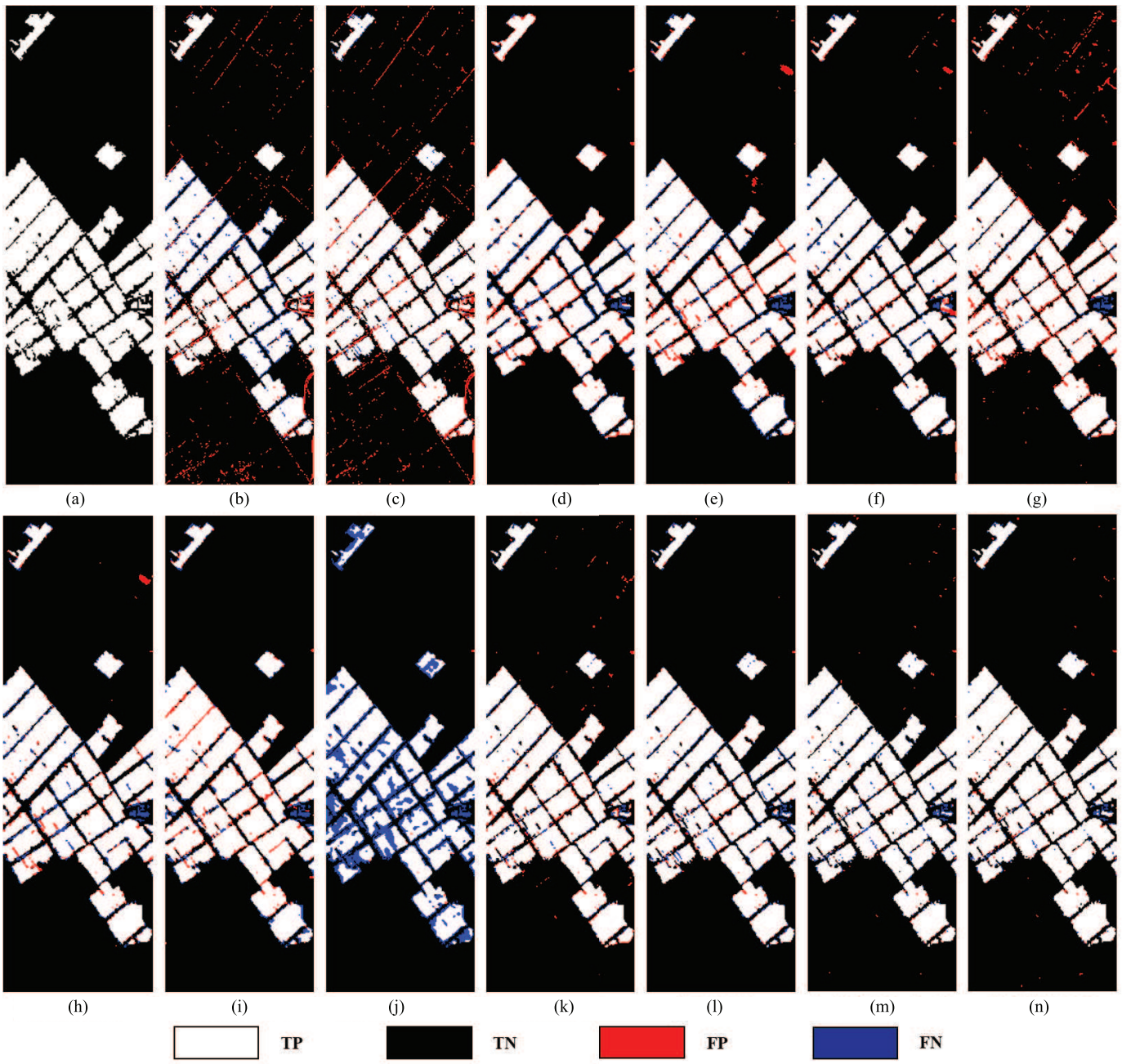



Fig. 7. CD maps of different methods on the Farmland dataset with 1% labeled training samples. (a) GT. (b) CVA. (c) DPCA. (d) CSANet. (e) SSTFormer. (f) MSDFFN. (g) DIEFEN. (h) HyGSTAN. (i) GLAFormer. (j) DADSSFF. (k) AIWSEN. (l) GASSM. (m) GDAMamba-p. (n) GDAMamba.

benchmark datasets. From top to bottom, the rows correspond to OA, Kappa, and $F1$-score, respectively. The experimental results demonstrate that GDAMamba consistently outperforms other models across varying training ratios. Notably, the performance variability among all the models diminishes as the number of training samples increases, suggesting enhanced feature extraction capability for changing regions when sufficient labeled data are available. However, under extremely limited training conditions (e.g., 0.1% sample ratio), GDAMamba exhibits significantly superior performance compared with other approaches, including its patchwise variant GDAMamba-P. This observation underscores the critical advantage of global image-level information utilization in maintaining model reliability under data scarcity conditions.

### B. Computational Complexity Analysis

Table II presents the statistics of the number of parameters (Params) and floating-point operations (FLOPs) for different models on the three benchmark datasets, offering a clear comparison of computational complexity. Thanks to Mamba's intrinsic capability of modeling global long-range dependencies with a minimal number of parameters, GDAMamba and its variant GDAMamba-P exhibit the lowest Params across all the datasets. They are followed by HyGSTAN, which adopts a lightweight design combined with spectral similarity filtering to reduce complexity. In addition, it can be observed that the patchwise GDAMamba-P exhibits the lowest FLOPs, while GDAMamba has the highest. This is attributed to GDAMamba's global contextual information extraction

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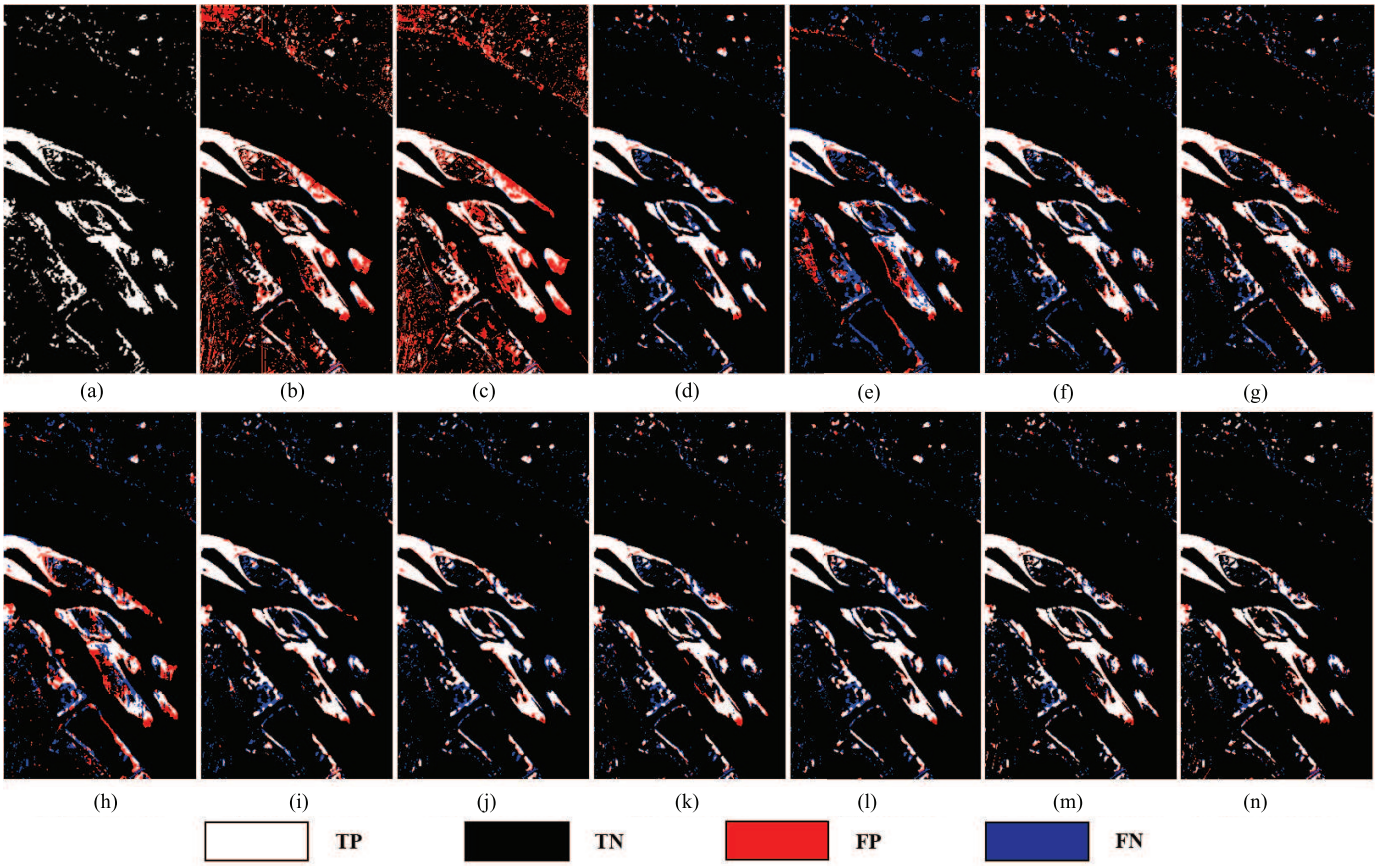



Fig. 8. CD maps of different methods on the river dataset with 1% labeled training samples. (a) GT. (b) CVA. (c) DPCA. (d) CSANet. (e) SSTFormer. (f) MSDFFN. (g) DIEFEN. (h) HyGSTAN. (i) GLAFormer. (j) DADSSFF. (k) AIWSEN. (l) GASSM. (m) GDAMamba-p. (n) GDAMamba.

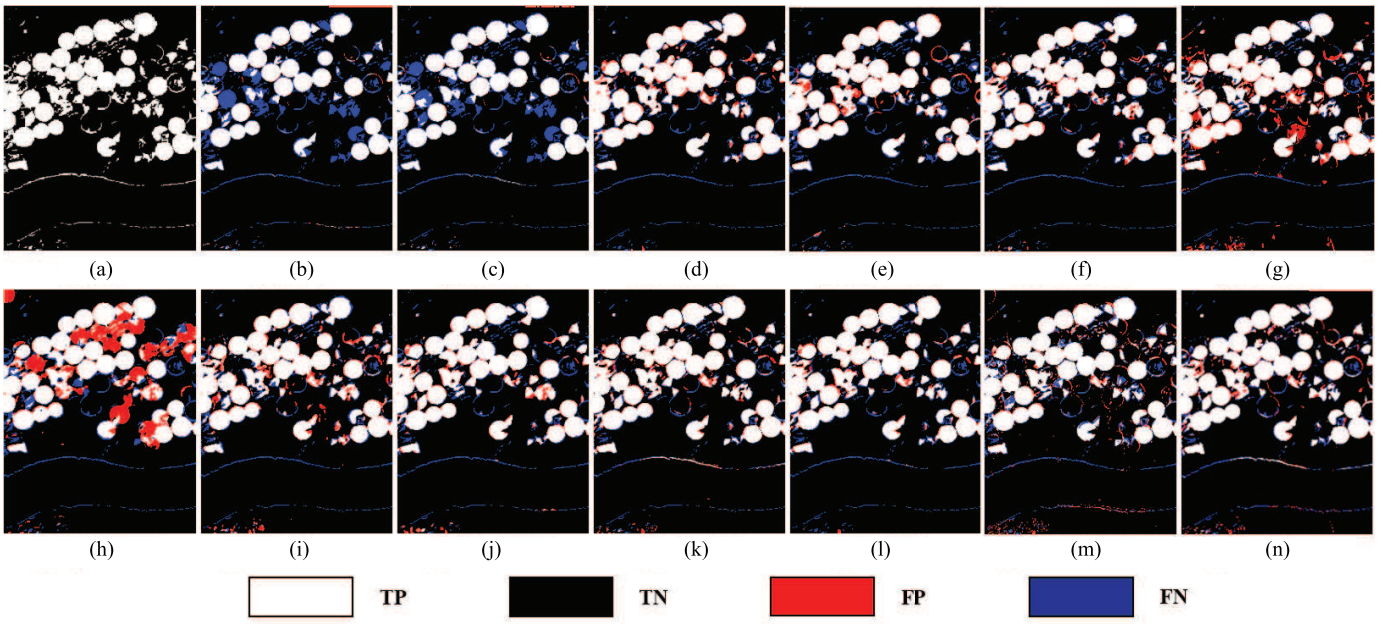



Fig. 9. CD maps of different methods on the USA dataset with 1% labeled training samples. (a) GT. (b) CVA. (c) DPCA. (d) CSANet. (e) SSTFormer. (f) MSDFFN. (g) DIEFEN. (h) HyGSTAN. (i) GLAFormer. (j) DADSSFF. (k) AIWSEN. (l) GASSM. (m) GDAMamba-p. (n) GDAMamba.

strategy, which models the relationships between each pixel and all others, thereby introducing a higher computational cost. Nevertheless, in terms of CD performance, this design achieves significantly better accuracy, demonstrating that the increase in FLOPs is justified by the substantial gains in detection effectiveness.

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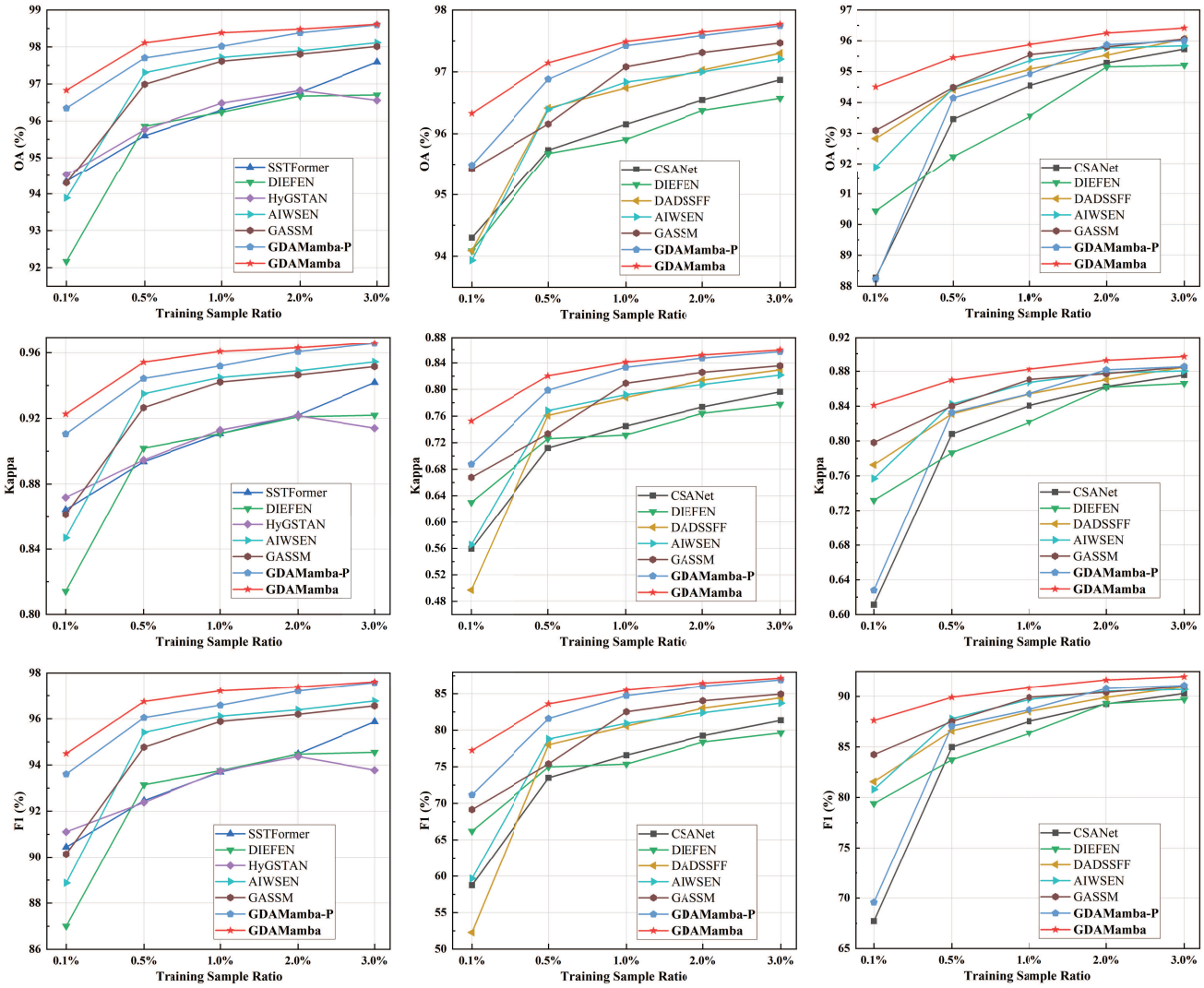



Fig. 10. CD performance (OA, Kappa, and $F1$-score) of different methods under varying training sample ratios on the three benchmark datasets. From top to bottom, the rows correspond to OA, Kappa, and $F1$-score, respectively. (a) Farmland. (b) River. (c) USA.

TABLE II
PARAMS AND FLOPS OF DIFFERENT MODELS ON THE THREE BENCHMARK DATASETS

| Datasets | Metrics | CSANet | SSTFormer | MSDFFN | DIEFEN | HyGSTAN | GLAFormer | DADSSFF | AIWSEN | GASSM | GDAMamba-P | GDAMamba |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Farmland | Params | 40.42M | 2.50M | 26.46M | 8.08M | 181.59K | 12.65M | 5.14M | 12.23M | 740.49K | **102.73K** | **102.73K** |
| | FLOPs | 287.39M | 320.16M | 1.96G | 42.39M | 8.87M | 1.93G | 261.04M | 124.82M | 53.04M | **4.51M** | 11.37G |
| River | Params | 40.44M | 2.53M | 26.46M | 8.22M | 282.9K | 12.67M | 5.71M | 12.50M | 1.05M | **105.48K** | **105.48K** |
| | FLOPs | 295.41M | 341.80M | 1.96G | 44.55M | 13.97M | 1.93G | 321.49M | 179.31M | 83.35M | **4.71M** | 21.05G |
| USA | Params | 40.42M | 2.50M | 24.46M | 8.08M | 179.5K | 12.65M | 5.13M | 12.23M | 734.18K | **102.66K** | **102.66K** |
| | FLOPs | 287.20M | 319.70M | 1.96G | 42.34M | 8.77M | 1.93G | 259.67M | 121.53M | 52.42M | **4.50M** | 13.33G |

Fig. 11 shows Params and the training time per epoch on the Farmland dataset. The Mamba-based GASSM, GDAMamba, and its variant GDAMamba-P consistently exhibit lower Params and reduced training time compared with transformer-based methods. GDAMamba-P presents a moderate training time due to its structural design that emphasizes capturing localized features and progressively enhancing pixelwise spectral differences across dual-temporal HSIs. In contrast, GDAMamba makes full use of hardware acceleration, treating each iteration as a complete epoch, thereby significantly reducing the time required per epoch.

### C. Ablation Study

Table III presents the statistical results of the ablation experiments. Here, GCDAM refers to the global contextual difference awareness module, and TSDEM denotes the temporal spectral difference enhancement module. In the case

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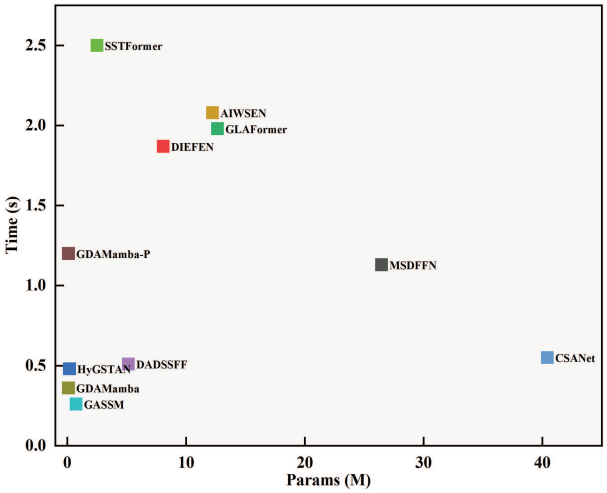



Fig. 11. Comparison results on model efficiency. The horizontal axis represents Params and the vertical axis represents the training time (time) for one epoch on the Farmland dataset.

TABLE III
ABLATION EXPERIMENTS OF GDAMAMBA ON THE THREE DATASETS

| Datasets | GCDAM | TSDEM | OA (%) | Kappa (×100) | F1 (%) |
|---|---|---|---|---|---|
| Farmland | × | × | 95.54±0.27 | 89.39±0.64 | 92.58±0.44 |
| | × | ✓ | 97.77±0.27 | 94.60±0.65 | 96.17±0.46 |
| | ✓ | × | 98.22±0.21 | 95.69±0.48 | 96.94±0.34 |
| | ✓ | ✓ | **98.38±0.17** | **96.08±0.41** | **97.22±0.29** |
| River | × | × | 89.46±4.12 | 55.93±11.0 | 61.08±9.30 |
| | × | ✓ | 97.13±0.19 | 80.34±1.46 | 81.89±1.36 |
| | ✓ | × | 97.44±0.12 | 83.68±0.93 | 85.08±0.87 |
| | ✓ | ✓ | **97.49±0.06** | **84.12±0.29** | **85.50±0.26** |
| USA | × | × | 93.53±0.34 | 81.82±0.74 | 86.02±0.51 |
| | × | ✓ | 95.65±0.16 | 87.24±0.58 | 90.02±0.50 |
| | ✓ | × | 95.78±0.32 | 87.96±0.84 | 90.69±0.63 |
| | ✓ | ✓ | **95.88±0.21** | **88.23±0.52** | **90.88±0.38** |

where neither module is used, a UNet architecture with two downsampling and upsampling layers serves as the baseline. When only TSDEM is used, its cascade structure generates multiple outputs, progressively amplifying feature differences layer by layer. These are then integrated into discriminative spectral features via the AGFM. It can be seen that training with 1% samples, high quantitative metrics have been achieved using the TSDEM-only strategy. This result confirms that gradually enhancing temporal spectral differences improves CD performance, highlighting the effective utilization of the intrinsic characteristics of HSIs. The GCDAM-only strategy also yields high-performance metrics, indicating that modeling global spectral differences enables the construction of associations between residual spectra at different locations, enhancing the model's perceptual and understanding capabilities at the image level. When the two modules are used simultaneously, performance improves further, indicating that TSDEM and GCDAM are complementary and stackable, jointly optimizing HSI-CD from both temporal enhancement and global context modeling perspectives.

### D. Impact of Parameter Settings

To further analyze the model architecture, we investigate the impact of different architectural settings on CD performance.

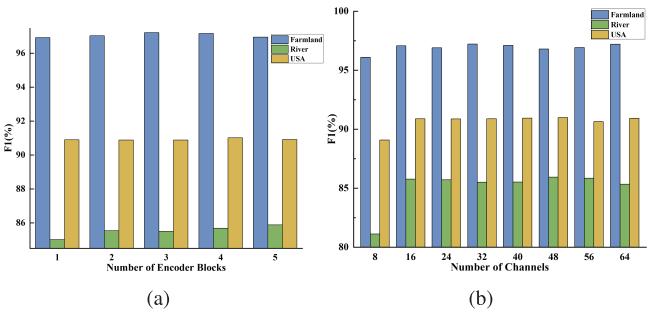



Fig. 12. $F1$-score performance of GDAMamba under different architectural settings. (a) Varying numbers of encoder blocks. (b) Varying feature channel dimensions.

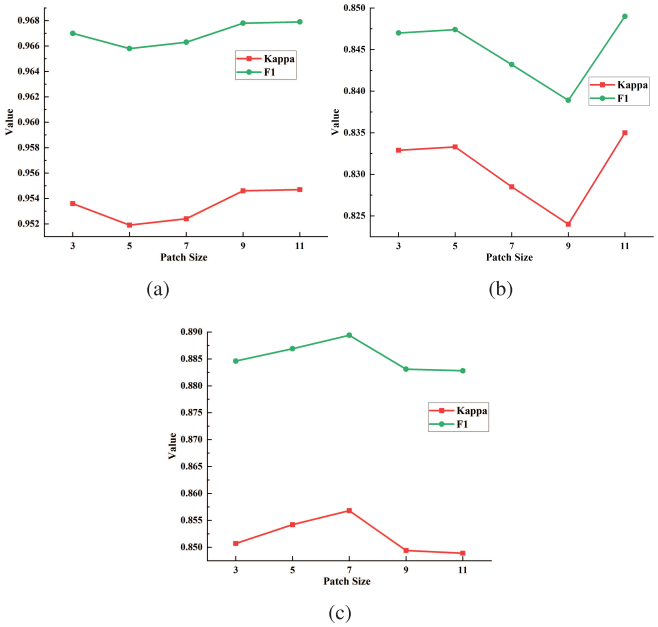



Fig. 13. Kappa and $F1$-score performance of GDAMamba-P with varying patch sizes. (a) Farmland. (b) River. (c) USA.

Fig. 12 illustrates CD performance under different combinations of encoder blocks and channel settings on the three benchmark datasets, using the $F1$-score as a representative metric. In terms of encoder blocks, the $F1$-score generally increases and then stabilizes across all the datasets. The optimal number of blocks is 3 for the Farmland dataset, 5 for the River dataset, and 4 for the USA dataset. Notably, the model maintains relatively stable performance across different block numbers, and even using a single encoder block without a deep architecture can outperform several more complex methods. This suggests that effectively leveraging global information can simplify network design while maintaining competitive performance. To prevent potential overfitting, a uniform setting of three blocks is adopted throughout this article. Regarding feature channels, the CD performance initially improves and then gradually stabilizes as the number of channels increases. After reaching 32 channels, performance differences across different settings become less pronounced. We uniformly set the number of feature channels to 32 for all the three datasets in this study.

The CD performance of GDAMamba-P under different patch sizes is further evaluated, with Kappa and $F1$-scores

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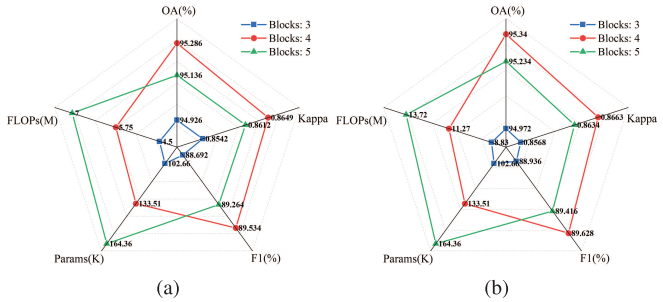



Fig. 14. In-depth analysis of GDAMamba-P on the USA dataset. (a) Patch size: $5 \times 5$. (b) Patch size: $7 \times 7$.

illustrated in Fig. 13. GDAMamba-P achieves the highest performance with a patch size of $11 \times 11$ on the Farmland and River datasets, while a patch size of $7 \times 7$ yields the best results on the USA dataset. These findings suggest that larger patches are more effective for scenes with broad and homogeneous change regions, whereas smaller patches are better suitable for areas with fragmented or complex changes, as in the USA dataset. Nevertheless, to balance accuracy and computational cost, we uniformly adopt a $5 \times 5$ patch size in this article. This moderate configuration enables GDAMamba-P to maintain strong CD accuracy while keeping the Params and FLOPs relatively low.

Fig. 14 provides a detailed analysis of the CD performance of GDAMamba-P on the USA dataset under different encoder block settings, using input patch sizes of $5 \times 5$ and $7 \times 7$. The best performance is achieved with four encoder blocks and a $7 \times 7$ patch size, reaching an OA of 95.286%, a Kappa coefficient of 0.8663, and an $F1$-score of 89.628%. Compared with the results reported in Table I, this configuration outperforms most existing methods and is comparable to the leading approaches. While this setting moderately increases Params and FLOPs, GDAMamba-P remains computationally efficient compared with other methods listed in Table II.

## V. Conclusion

In this article, we propose GDAMamba, the first Mamba-based image-level method for HSI-CD, which consists of three key modules: GCDAM, TSDEM, and AGFM. The experimental results on three benchmark datasets demonstrate that GDAMamba achieves state-of-the-art performance under various training sample ratios and remains highly reliable even with very limited training data, highlighting its robustness and effectiveness. The ablation study and quantitative evaluations verify that modeling the association between each pixel and all others in the difference map via global difference awareness significantly improves CD accuracy. In addition, enhancing temporal spectral differences through TSDEM further contributes to performance gains. As the first fully Mamba-based image-level approach for HSI-CD, GDAMamba presents a novel and effective pathway for advancing research in this field.

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